Sneha Patra **D20A** **43**

Blockchain Experiment 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

THEORY:

Q1: What is MetaMask?

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with blockchain networks such as Ethereum. It enables users to store, send, and receive digital assets securely. MetaMask also acts as a bridge between web browsers and decentralized applications (DApps), allowing users to sign transactions and deploy smart contracts. It provides account management, private key security, and seamless integration with platforms like Remix IDE.

Q2: What is a Test Network (Testnet)?

A test network, or testnet, is a blockchain environment used for testing purposes. It simulates the functionality of the main Ethereum network but uses virtual currency instead of real funds. Developers use testnets like Sepolia or Goerli to deploy and test smart contracts without financial risk. Testnets help identify bugs, verify functionality, and ensure proper execution before deploying applications on the mainnet.

Q3: Steps to Connect MetaMask with Remix IDE for Performing Transactions

- Install the MetaMask extension from the Chrome Web Store
- Create a new wallet or import an existing wallet using a secret recovery phrase
- Enable test networks in MetaMask settings
- Select a test network such as Sepolia
- Add test ETH using a faucet for performing transactions
- Open Remix IDE in the browser (https://remix.ethereum.org)
- Write and compile the smart contract in Remix
- Go to the Deploy & Run Transactions tab
- Select Injected Provider - MetaMask as the environment
- Connect MetaMask account when prompted
- Click Deploy and confirm the transaction in MetaMask
- Interact with deployed contract functions and approve transactions via MetaMask

CODE:

Smart Contract 1: OrderRecord.sol

Records each order placed on the platform permanently on the blockchain. Stores the buyer's wallet address, a comma-separated list of product IDs, the total amount in paisa, and a timestamp.

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.0;
contract OrderRecord {
    struct Order {
        address buyer;
        string productIds;    // comma-separated e.g. "p1,p2,p3"
        uint256 totalPaisa;   // ₹199.99 stored as 19999
        uint256 timestamp;
    }

    Order[] public orders;

    event OrderPlaced(address buyer, string productIds, uint256
totalPaisa, uint256 timestamp);

    function placeOrder(string memory _productIds, uint256 _totalPaisa)
public {
        orders.push(Order(msg.sender, _productIds, _totalPaisa,
block.timestamp));
        emit OrderPlaced(msg.sender, _productIds, _totalPaisa,
block.timestamp);
    }

    function getOrderCount() public view returns (uint256) {
        return orders.length;
    }
}
```

Smart Contract 2: ReviewVerification.sol

Records product reviews on the blockchain to ensure authenticity and prevent tampering. Stores the product ID, reviewer's wallet address, star rating (1–5), and a timestamp. Also prevents the same wallet from reviewing the same product twice.

```
//Smart Contract 2: Review Verification.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract ReviewVerification {
    struct Review {
        string productId;
```

```
        address reviewer;
        uint8 rating;
        uint256 timestamp;
    }

    Review[] public reviews;
    mapping(string => mapping(address => bool)) public hasReviewed;

    event ReviewAdded(string productId, address reviewer, uint8 rating,
uint256 timestamp);

    function addReview(string memory _productId, uint8 _rating) public {
        require(_rating >= 1 && _rating <= 5, "Rating must be 1-5");
        require(!hasReviewed[_productId][msg.sender], "Already reviewed");

        reviews.push(Review(_productId, msg.sender, _rating,
block.timestamp));
        hasReviewed[_productId][msg.sender] = true;

        emit ReviewAdded(_productId, msg.sender, _rating,
block.timestamp);
    }

    function getReviewCount() public view returns (uint256) {
        return reviews.length;
    }
}
```

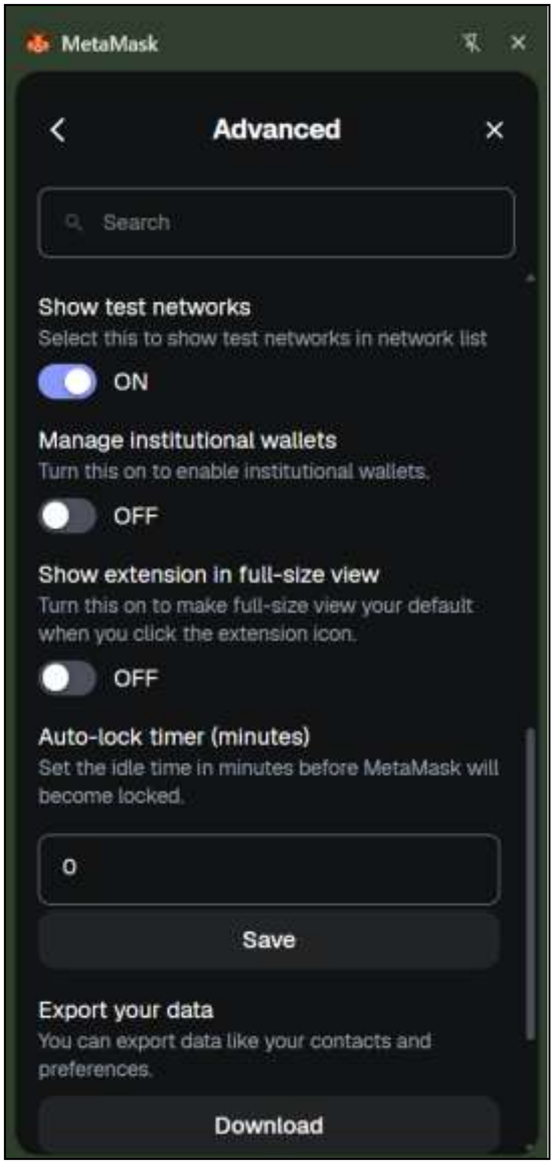
PROCEDURE

Step 1: Install Metamask Chrome extension

Install the MetaMask wallet extension from the Chrome Web Store. MetaMask is used to interact with the Ethereum blockchain and manage Ethereum accounts.
LINK:
https://chromewebstore.google.com/detail/metamask/nkbihfbeogaeaoehlefnkodbefgpgknn?pli=1

Step 2: Enable Test Network

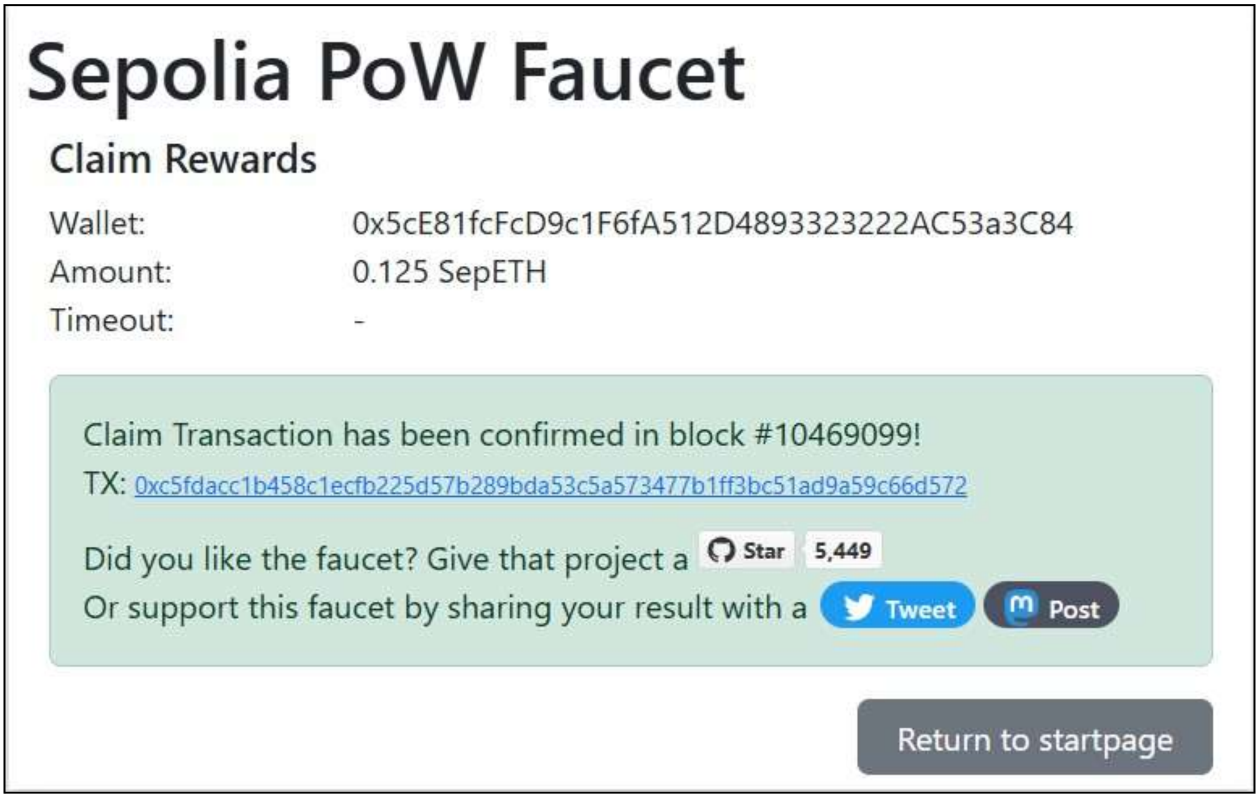
Open MetaMask and enable Test Networks from the settings. Select the Sepolia Test Network which allows users to test smart contracts without using real cryptocurrency.



Step 3: Claim free test ETH

Visit the Sepolia Faucet website and request free test Ethereum (ETH). These tokens are used to deploy and test smart contracts on the Sepolia test network.

https://sepolia-faucet.pk910.de/#/claim/c21e4c9c-4717-417f-b7cb-7fb785414d23



Step 4: Open Remix IDE

Open the Remix Integrated Development Environment in the browser. Remix IDE is used to write, compile, deploy, and test Solidity smart contracts.

https://remix.ethereum.org

Step 5: Create Smart contracts

a. **Order Record Contract**

   Records every order placed through the application permanently on the blockchain.

   Stores on blockchain: buyer wallet address, product IDs (comma-separated), total amount in paisa, and timestamp.

   - buyer wallet address
   - product IDs (comma-separated)
   - total amount in paisa
   - timestamp.

   Benefit:

   - Prevents tampering of order records
   - Provides transparent and immutable proof of purchase

b. **Review Verification Contract**

   Ensures that product reviews are genuine and cannot be altered after submission.

   Stores on blockchain:

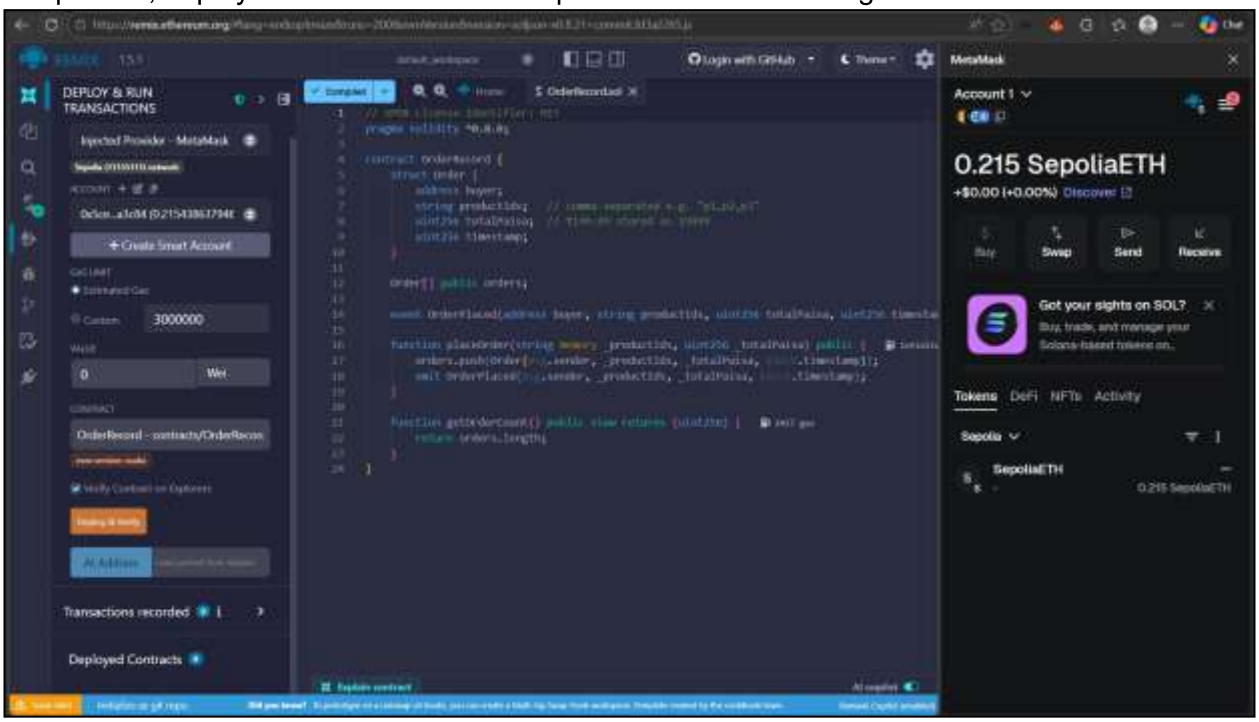
- product ID
- reviewer wallet address
- rating (1–5)
- timestamp.

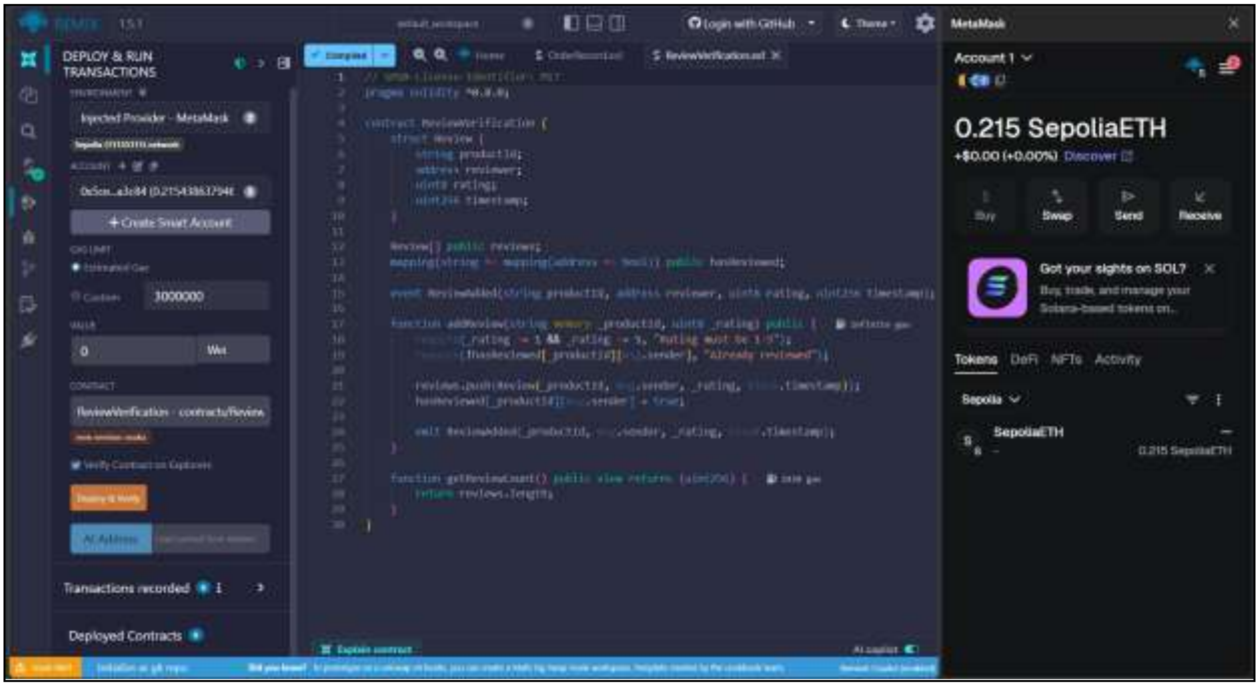
Benefit:

- Prevents fake or duplicate reviews from the same wallet
- Provides immutable, verifiable proof of each review

Step 6: Compile and deploy Smart contracts

Compile the smart contracts using the Solidity compiler in Remix IDE. After successful compilation, deploy the contracts on the Sepolia Test Network using the MetaMask wallet.

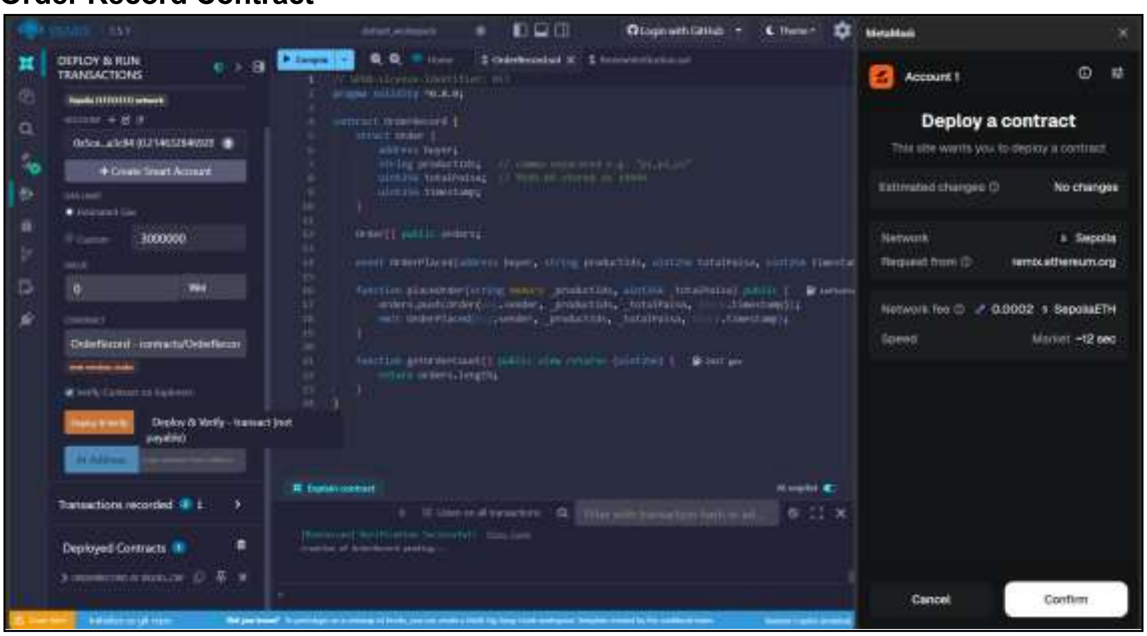


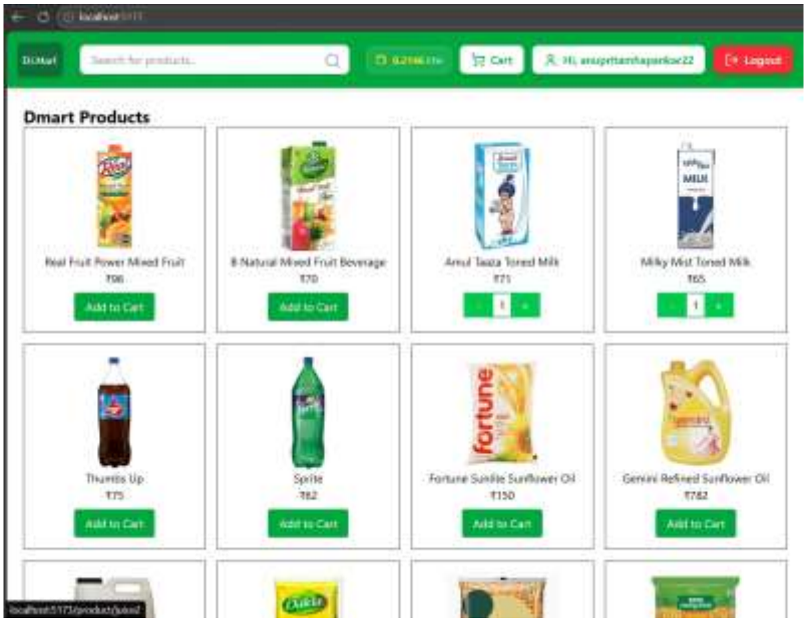


Step 7: Test the contracts

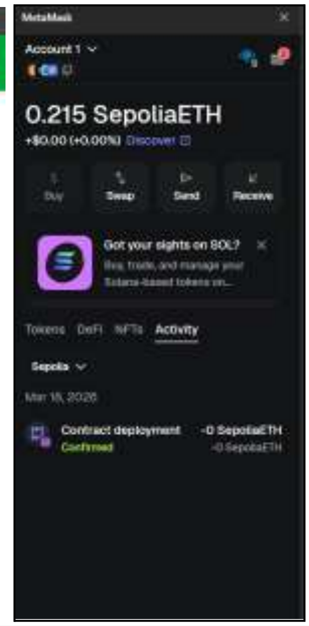
After deployment, test the contract functions such as addProduct, getProduct, and addReview. Verify that product details and reviews are stored correctly on the blockchain and can be retrieved using the contract functions.

a. **Order Record Contract**

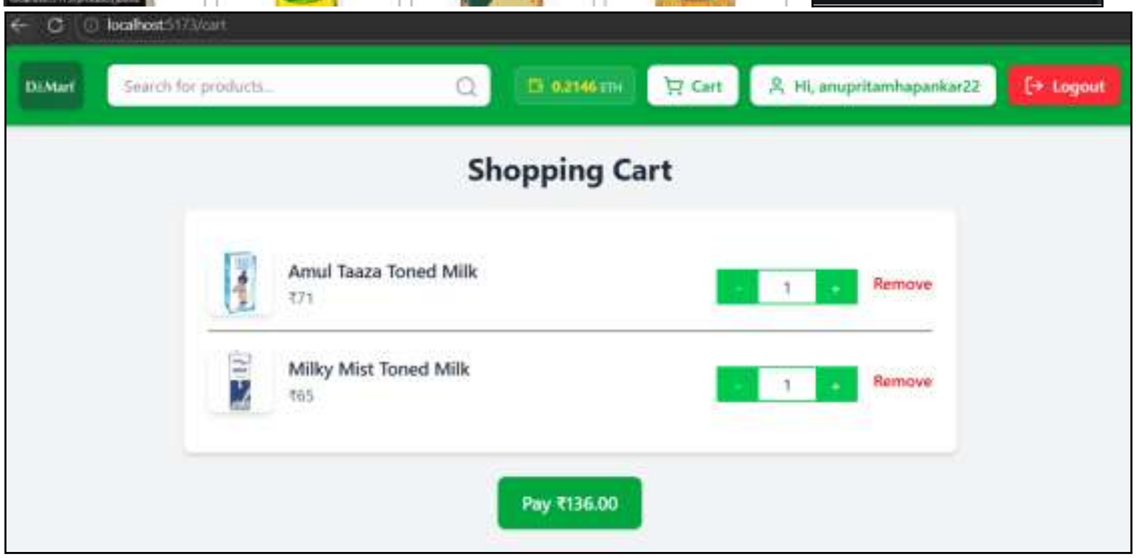




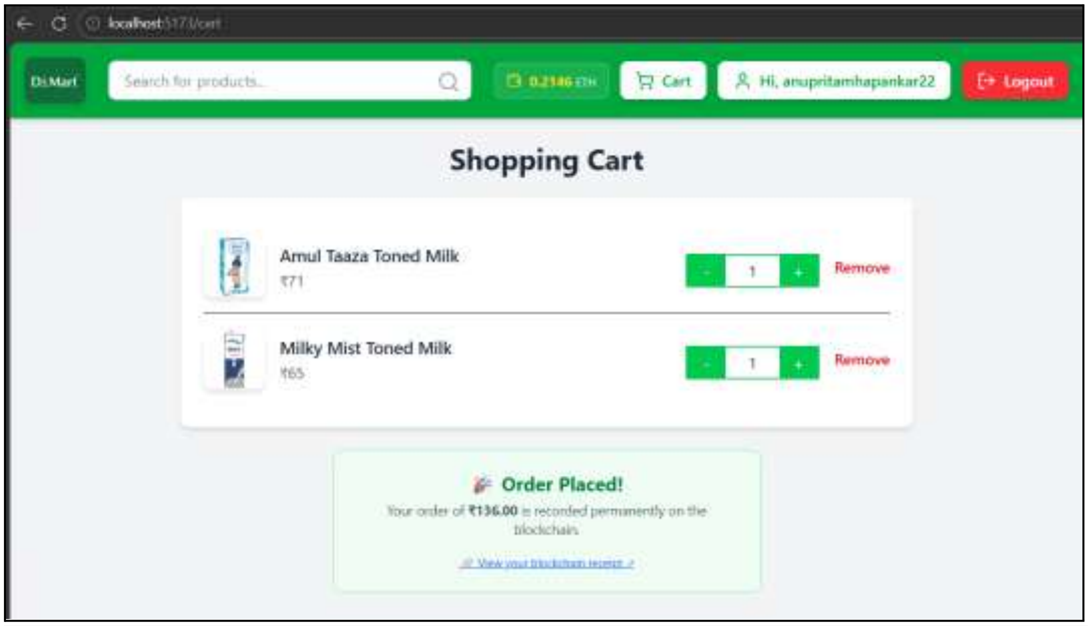
Dmart Products
Real Fruit Power Mixed Fruit
B Natural Mixed Fruit Beverage
Amul Taaza Toned Milk
Milky Mist Toned Milk
Add to Cart
Thums Up
Sprite
Fortune Sunlite Sunflower Oil
Gemini Refined Sunflower Oil
Logout

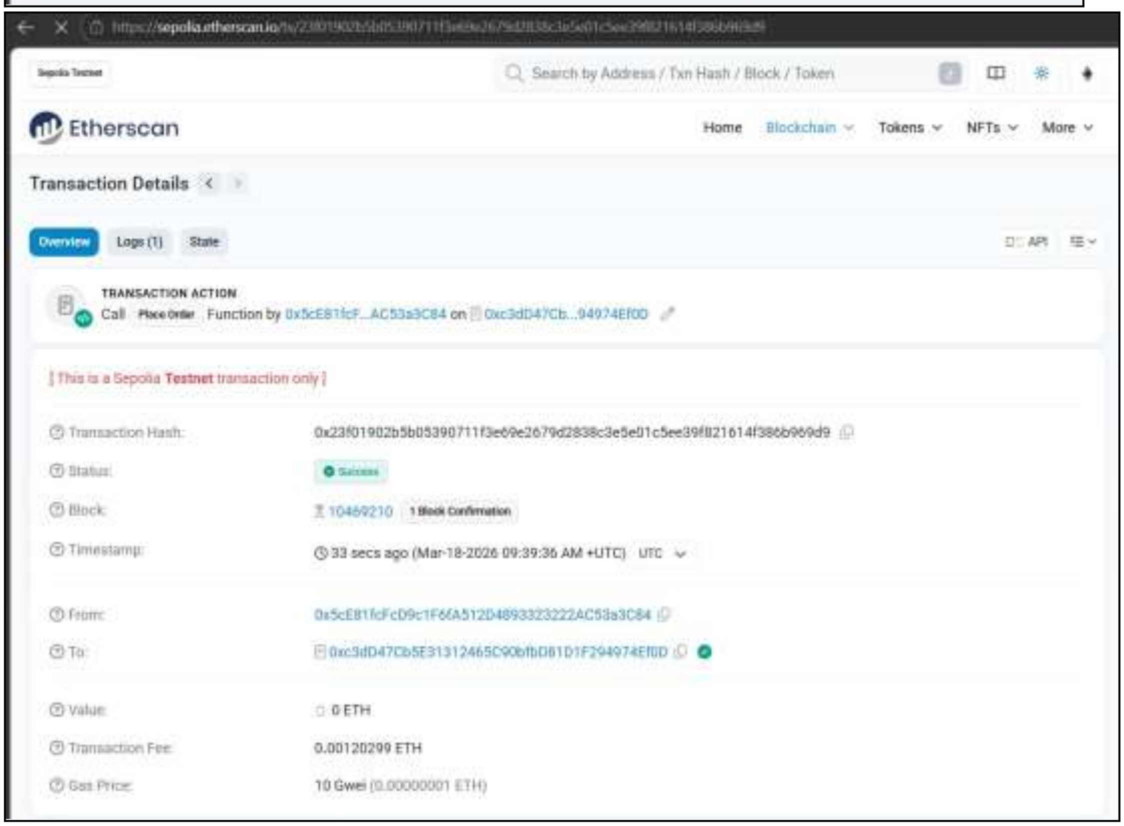


MetaMask
Account 1
0.215 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Got your sights on SOL?
Tokens
DeFi
NFTs
Activity
Sepolia
Contract deployment
Confirmed
-0 SepoliaETH

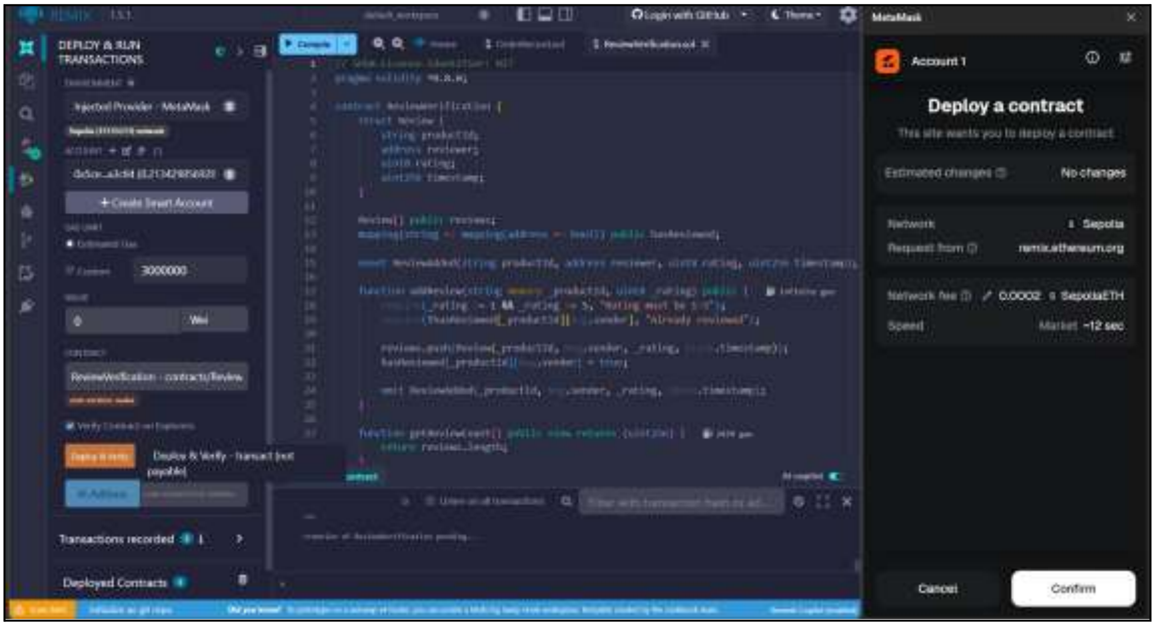


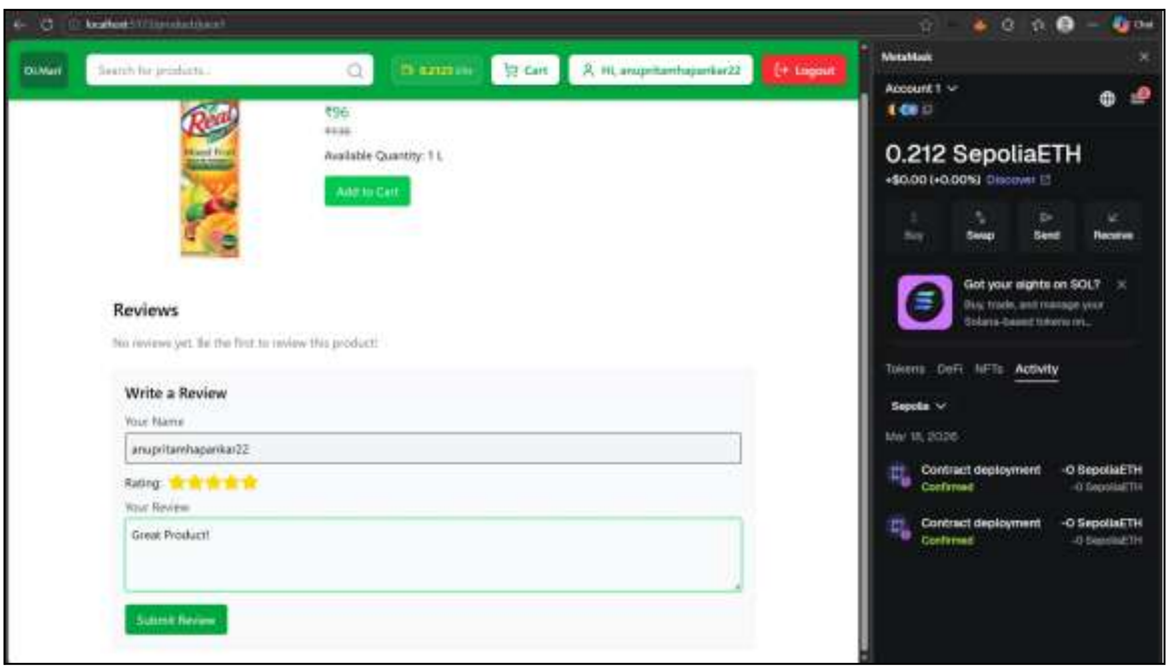
localhost:5173/cart
DMart
Search for products...
0.2146 ETH
Cart
Hi, anupritamhapankar22
Logout
Shopping Cart
Amul Taaza Toned Milk
₹71
Remove
Milky Mist Toned Milk
₹65
Remove
Pay ₹136.00

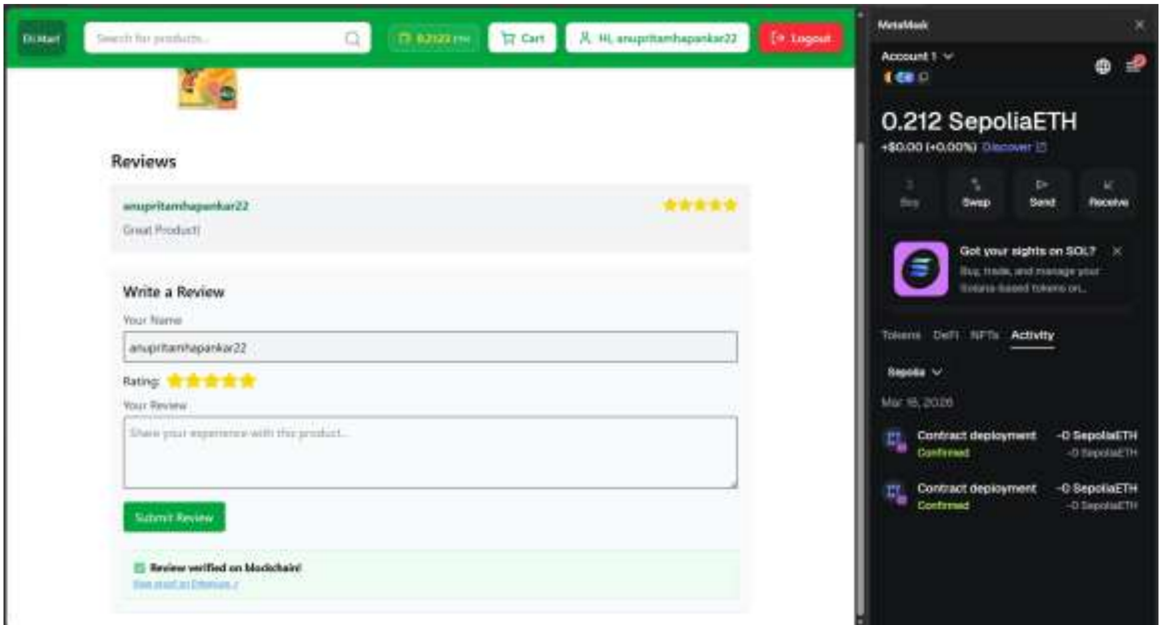

DMart
Search for products...
0.2146 ETH
Cart
Hi, anupritamhapankar22
Logout
Shopping Cart
Amul Taaza Toned Milk
₹71
Remove
Milky Mist Toned Milk
₹65
Remove
Order Placed!
Your order of ₹136.00 is recorded permanently on the blockchain.
View your blockchain receipt

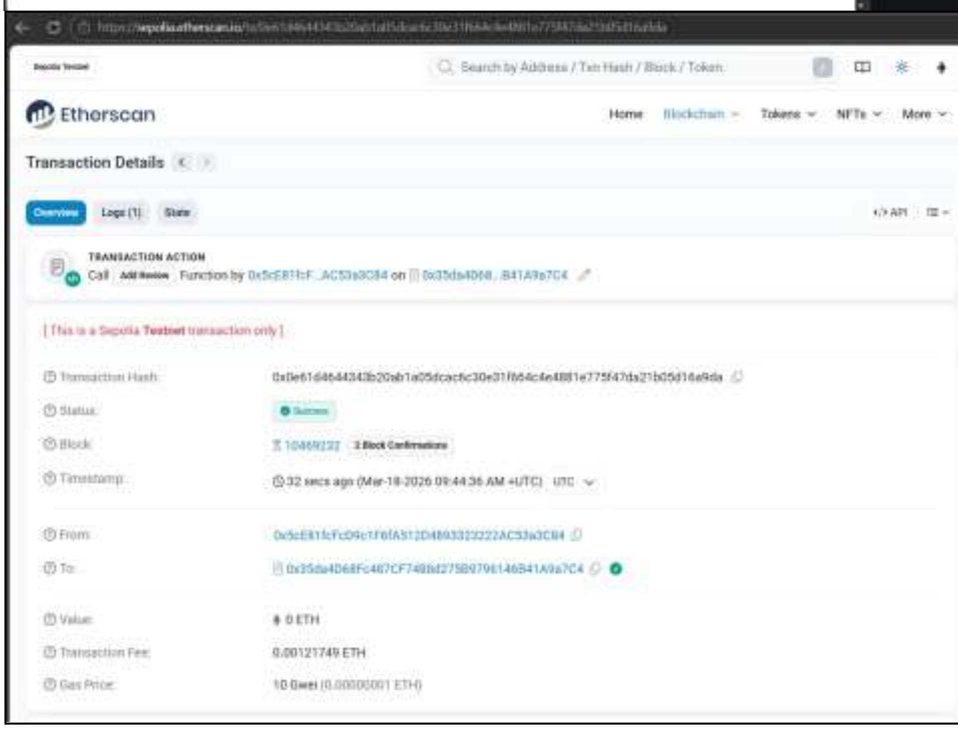

Sepolia Testnet
Search by Address / Txn Hash / Block / Token
Etherscan
Home
Blockchain
Tokens
NFTs
More
Transaction Details
Overview
Logs (1)
State
API
TRANSACTION ACTION
Call Place Order Function by 0x5cE81fcF...AC53a3C84 on 0xc3dD47Cb...94974Ef0D
[This is a Sepolia Testnet transaction only]
Transaction Hash:
0x23f01902b5b05390711f3e69e2679d2838c3e5e01c5ee39f821614f386b969d9
Status:
Success
Block:
10469210
1 Block Confirmation
Timestamp:
33 secs ago (Mar-18-2026 09:39:36 AM +UTC)
UTC
From:
0x5cE81fcFcD9c1F6fA512D4893323222AC53a3C84
To:
0xc3dD47Cb5E31312465C90bfbD81D1F294974Ef0D
Value:
0 ETH
Transaction Fee:
0.00120299 ETH
Gas Price:
10 Gwei (0.00000001 ETH)

b. **Review Verification Contract**









CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing, compiling, deploying, and testing smart contracts on the Ethereum blockchain.